

NewsBot Intelligence System 2.0 - Reflective Journal

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Why did I choose this. I wanted something I wouldn't be ashamed of presenting in front of a non-technical audience. This means minimal installation, expected outcomes, and explanatory artifacts - plots, tables, and CSVs that support every assertion.

What surprised me. There weren't always labels in the dataset. It may seem minor, but it altered the entire project. I had to create labels from nothing and still ensure the split was stratified and the rest of the pipeline was neat. It really seemed like an industry problem where the data isn't exactly how you'd like it.

What did you find worked best? Simple things are done well. It's hard to beat TF-IDF with a linear classifier if you want an accurate, interpretable baseline. NMF and LDA showed different sides of the same information. VADER wouldn't necessarily win a conference award, but it's good enough for this domain. A small Gradio app softened the testing experience.

What I'd do with additional time. Use embeddings for search, a small abstractive summarizer on a sample, named-entity recognition, and change of topic over time. That would bring it closer to something useful for a newsroom.

What I took away from it. It's just as beneficial to bring together so many diverse pieces into one integrated effort as it is to enjoy/focus/accomplish well with any one tangible quality. Making it replicable and valuable is a lesson unto itself. Therefore, it's an appreciation for every minutia worked both in my favor and against me for completing it all on my own because at the moment it's stressful, but in the end, it's all worth it.